THE STRUCTURE PUSHFORWARD OF A PERFECT CLOSED IMMERSION IS A PERFECT COMPLEX

THE STACKS PROJECT AUTHORS

Editorial note. Statement span is lines 17953-17958 of sources/stacks-more-morphisms/pinned-original-source.tex; proof span is lines 17960-17970, the complete author proof (9 body lines).

Editorial note. Navigation stubs: 2 cited-but-not-included labels are supplied as non-load-bearing pointers with their pinned tags where the snapshot tag index has them.

Editorial note. extra_packages is empty on purpose: the retained preamble already loads xy with options and a second declaration clashes in the assembler wrapper.

Editorial note. Text is retained verbatim, including the author's own typos and placeholders.

Editorial note. Module-level companion of the morphism-level perfection of regular immersions (L16121); both feed the chapter's perfect-pushforward calculus.

Allowed standard prerequisites (editorial).

- Explicit prerequisite boundary. Definition of perfect closed immersion (locally cut out by Koszul-regular sequences, chapter definition section); More on Algebra, lemma perfect-ring-map (finite projective resolution from a perfect ring map); Cohomology, section perfect (the object-level definition being verified).

Source reference. Original lemma 08CL: sources/stacks-more-morphisms/pinned-original-source.tex, source lines 17970–17970. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

Source reference. Original lemma 068Y: sources/stacks-more-morphisms/pinned-original-source.tex, source lines 17970–17970. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

Lemma 0.1. *Let $i : Z \to X$ be a perfect closed immersion of schemes. Then $i_*\mathcal{O}_Z$ is a perfect $\mathcal{O}_X$-module, i.e., it is a perfect object of $D(\mathcal{O}_X)$.*

Proof. This is more or less immediate from the definition. Namely, let $U = \mathrm{Spec}(A)$ be an affine open of X. Then $i^{-1}(U) = \mathrm{Spec}(A/I)$ for some ideal $I \subset A$ and A/I has a finite resolution by finite projective A-modules by More on Algebra, Lemma 068Y. Hence $i_*\mathcal{O}_Z|_U$ can be represented by a finite length complex of finite locally free $\mathcal{O}_U$-modules. This is what we had to show, see Cohomology, Section 08CL. □

Date: stacks-project:more-morphisms.tex:local-snapshot-20261011; GNU FDL 1.2 or later.